

ECE 2195: Engineering with AI

Homework 2 - Prompt design as engineering specification

100 points (+10pts extra credit)

Due October 2 @11:59pm, on Canvas

Objective: Advance from informal prompting toward formal specification design.

Other instructions: *Work in groups of 2-3. You are allowed (and encouraged!) to get help from AI tools for this HW assignment. After the HW deadline, come prepared to class to discuss your experiments, outcomes, and what you learned from this HW.*

Assignment:

Ask an AI system to design a simple data-acquisition system that samples a temperature sensor and reports abnormal measurements.

Part A — Baseline Prompt

1. Begin with a simple natural-language request.
2. Evaluate the resulting response for:
 - completeness
 - correctness
 - specification compliance

Part B — Structured Prompt

1. Redesign the prompt using explicit components:
 - system/context
 - engineering objective
 - assumptions
 - input information
 - constraints
 - required reasoning/task steps
 - output requirements

For example:

```
SYSTEM CONTEXT
You are assisting with the design of an embedded monitoring system.

TASK
Design a temperature acquisition subsystem.

REQUIREMENTS
Sampling rate: 10 Hz
ADC resolution: 12 bits
Supply voltage: 3.3 V
...

OUTPUT
Return:
1. Architecture
2. Required components
3. Fault detection strategy
4. Design assumptions
```

Part C — Few-shot Design

1. Compare:
 - zero-shot
 - one-shot
 - three-shot
2. Justify **why you selected these examples** (the examples shouldn't be inserted arbitrarily).

Part D — Structured Output

1. Require the model to return a machine-readable result such as:

```
{
  "architecture": [],
  "components": [],
  "fault_detection": [],
  "assumptions": [],
  "risks": []
}
```
2. Test whether the output complies with the schema.

Deliverables

1. A short report containing:
 - baseline prompt
 - structured prompt
 - few-shot variants
 - comparison table
 - recommended prompt architecture
 - design rationale
2. Any code or scripts used

Extra credit (10 points): Conduct a small **example-selection experiment** comparing different few-shot example sets or orderings and determine whether example choice significantly affects performance.